

MILESTONE 4

1. Overview:

This website is a modern web application designed to simplify and accelerate the medical diagnosis process, specifically for analyzing X-ray images. The platform allows doctors and patients to interact with a trained AI model to predict lung diseases quickly and accurately. The system provides a user-friendly interface and real-time predictions, making it an efficient tool for medical professionals and patients. The website is developed using modern web technologies:

Front-end: React with Vite, providing a fast, responsive, and interactive user interface.

Back-end: Python Flask, handling image processing, communication with the AI model, and user management.

2. User System:

The website includes a login system with two types of users:

Doctors: Can upload patient X-ray images, view detailed results, and generate reports.

Patients: Can upload their own X-ray images to receive predictions and view their diagnostic reports.

3. Key Features:

The website provides a wide range of features designed to enhance usability and functionality:

Image Upload and Processing

Users can upload X-ray images for analysis.

Images are preprocessed to improve prediction accuracy.

AI-Based Diagnosis

The uploaded images are passed through a deep learning model.

The system predicts lung diseases and generates visualizations for easy interpretation.

Real-Time Results

Users receive predictions instantly.

Detailed metrics are provided, including accuracy, precision, recall, and F1 score.

Report Generation

Users can download results as a PDF report for record-keeping or sharing.

User Interface Enhancements

Multi-language support: English, Telugu, Hindi, Kannada, Tamil.

Dark and light modes for a comfortable viewing experience.

Interactive dashboards showing results and visualizations clearly.

AI Chatbot Assistance

A 24/7 AI chatbot is available to answer user queries and guide them through the process.

4. Challenges and Solutions

Building this system involved addressing several challenges:

Low-quality X-ray images

Imbalanced datasets

Similarities between different lung diseases

Solutions implemented:

Structured workflow with preprocessing steps to improve image quality.

Advanced deep learning techniques to enhance model accuracy.

Visualization outputs to clearly distinguish predictions.

5. Workflow:

The process for diagnosis follows a structured workflow:

Image Upload: Users upload X-ray images.

Preprocessing: Images are enhanced and standardized.

Prediction: The AI model analyzes the image and predicts the disease.

Visualization: Results are displayed with clear visualizations.

Report Generation: Users can download reports including prediction scores.

6. Strengths:

High accuracy and precision in predictions.

Real-time predictions for fast diagnosis.

User-friendly interface for doctors and patients.

Multi-language support and customizable display modes.

Comprehensive reporting with exportable PDFs.

24/7 AI chatbot assistance.

Deploy link: <https://cliniscan-frontend.vercel.app/>